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An Efficient Industrial Robot Calibrator With Multiplaner Constraints

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***Abstract*—Calibration technology has become an essential part of the task of improving robot precision. However, existing calibration schemes suffer from accuracy loss in actual working conditions due to the ignorance of the target robot's constrained workspace. In response to this challenge, this study proposes a novel robot calibrator with three-fold ideas: 1) optimizing the measurement configuration via the measurement configurations selection (MCS) for suppressing the measurement noises, 2) developing the alternation-direction-method-of-multipliers with multiplanar constraints (AMPC) algorithm for superior calibration accuracy and mitigatory long-tail convergence, with its convergence being theoretically proved, and 3) building the MCS-AMPC-based robot calibrator for efficient kinematic parameters calibration. For validating its performance, the HRS-P dataset that contains 2400 samples collected from different working spaces of an HRS-JR680 industrial robot has been established and publicly released. Empirical studies conclusively affirm that, compared to the Levenberg–Marquardt algorithm (the most accurate among several state-of-the-art algorithms), the proposed AMPC algorithm reduces the mean error by 10.34%. After denoising the calibration data with MCS and then employing the AMPC algorithm to identify the robot's kinematic parameters, the calibration accuracy can be further improved. Moreover, building an ensemble to address the issue of various local optimal solutions can also contribute to better calibration accuracy. The empirical results highlight the superiority and scalability of the MCS-AMPC calibrator for industrial robot-related applications.**

***Index Terms*—Alternation direction method of multipliers, calibrator, industrial robot calibration, kinematic parameters, measurement configurations selection, multiplanar constraints.**

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I. Introduction

NOWADAYS, advanced and comprehensive industrial robot technology plays a pivotal role in driving the development level of high-end manufacturing. With the increasingly widespread application of industrial robots in various fields, including industrial product manufacturing, logistics sorting, food processing, aerospace science and engineering, and precision medical treatment the demand for higher precision has grown significantly [1], [2], [3]. It is worth highlighting that, although most industrial robots have high repeatability accuracy, their absolute positioning accuracy still exhibits a relatively lower level [2]. Therefore, it is essential to perform periodic calibrations, and the factors influencing the absolute positioning accuracy are generally classified into non-geometric and geometric errors [3], [4]. Notably, 90% of the total error source comes from geometric errors, and this study mainly addresses this error issue in the calibration process [4].

Kinematic parameter calibration [1], [2], [3], [4], [5], [6], a powerful and effective technique, can reduce industrial robots' geometric errors by adjusting their joint angles and link lengths. Currently, numerous researchers have conducted extensive studies on this technique. Liu et al. [3] presented a novel scheme for robot calibration with a six-point measurement device and identify the kinematic parameters with a Levenberg–Marquardt (LM) algorithm, which successfully mitigates the maximum error of a UR10 robot. Chen and Zhan [5] proposed a beetle swarm optimization (BSO) algorithm based on an integration of particle swarm optimization (PSO) and beetle antennae search algorithms and demonstrate its calibration performance with empirical studies. Body et al. [7] measured the workspace of the Fanuc 165F robot using a monocular camera with fiducial markers, and utilize a least-square (LS) algorithm to optimize the solution model, which achieves a 45% reduction in position error. Cao et al. [9] provided a butterfly and flower pollination algorithm to improve an artificial neural network algorithm. Experiments conducted on a Stewart platform demonstrate its outstanding performance.

Note that the above methods are only applicable when the robot's workspace is intact and the data samples can cover the whole sample space. Besides, industrial robots are frequently restricted by the actual working environment, e.g., walls or other equipment, leading to a shortage of samples during the calibration process [10], [11], [12], [13]. To address this tough issue, Scholars specializing in robot calibration have devised diverse methods to collect samples within constrained

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spaces [10], [11], [12], [13], [14], [15]. Lembono et al. [10] introduced two spatial constraints, i.e., a planar constraint and a distance constraint, and utilize the laser and the tooltip to enable robot calibration with an optimal LM algorithm. Wang et al. [12] collected datasets from a 6R robot under point and distance constraints and employ the LM algorithm for their training. Joubair and Bonev [13] incorporated the planar constraints into the Gauss–Newton algorithm to calibrate a six-DOF FANUC LR Mate 200iC robot, which mitigates its maximum error. Stepanova et al. [14] presented a complementary robot calibration scheme based on the self-observation-contact and planar constraints and calibrate kinematic parameters with the LM and Gauss–Newton algorithms. However, these methods commonly rely on traditional algorithms based on the Jacobian matrix, i.e., the LM [10], [12], [14], the Gauss–Newton [13], [14] and the LS [9] algorithms, which frequently suffer from low convergence rates, long-tail convergence, and local optimum, resulting in low computational efficiency [4]. Motivated by these discoveries, this study proposes a novel and efficient alternation-direction-method-of-multipliers with multiplanar constraints (AMPC) algorithm, which contains two-fold ideas.

1) Establishing an alternation-direction-method-of-multipliers (ADMM) solution model to facilitate rapid convergence in the robot kinematic parameter calibration process.
2) Integrating multiplanar constraints (MPC) with the ADMM algorithm to address the low accuracy and robustness issues caused by limited sample range.

Moreover, suppressing measurement noise is another crucial step during the calibration procedure [16], [17], [18], which can rely on the optimized measurement configuration, and employs observability indices as the evaluation metric. Specifically, the measurement configuration selection is based on the singular value decomposition (SVD) method, which can filter out actual measurement noise by retaining relatively large singular values to reconstruct the original data. In general, a higher observability index corresponds to lower measurement noise, bringing the samples closer to the true values and ensuring stable calibration performance [16]. Thus, this article adopts a differential evolution (DE) algorithm [1] based on the observability index to optimize the process of the measurement configurations selection (MCS), aiming to minimize the impact of the measurement noise on parameter identification.

Hence, the main contributions of this article include the following.

1) A public MPC-based dataset HSR-P for the robot calibration. Each sample includes a drawstring length, a dial reading and six joint angles, and it is available on the website[1] to provide data assistance for practitioners.
2) An AMPC algorithm. It combines the MPC with the ADMM algorithm, ensuring both convergence rate and calibration accuracy, with its theoretical convergence conclusively proven.
3) An MCS-AMPC calibrator. It achieves superior calibration accuracy and computational efficiency. Besides, the algorithm design and analysis are comprehensively summarized, serving as a valuable reference for researchers and engineers addressing robot calibration challenges.
4) An ensemble. It is constructed from several local optimal solutions, resulting in the final solution, which can contribute to better calibration accuracy.

The rest of this article is organized as follows. Section II presents the preliminaries. Section III describes the MCS-AMPC calibrators for kinematic parameter identification in detail. The experimental results are reported in Section IV. Finally, Section V concludes this article. The pseudocode for MCS-AMPC, the detailed convergence proof for AMCP, and part of the tables and figures related to notations and experiments are left in Supplementary File (SF)[2].

II. Preliminaries

This section mainly discusses the robot kinematic model and the robot kinematic model error. Table S.I lists the involved symbols in this article.

A. Robot Kinematic and Error Models

The kinematics calibration technology for industrial robots generally involves four primary steps, i.e., modeling, measuring, parameter identifying, and error compensating. The D-H model (presented by Denavit and Hartenberg) is the most prevalent one in robot modeling, and its D-H parameters can be directly output by the majority of robotic teach pendants [4]. Thus, we adopt this model to describe the basic conversion relationship between adjacent joint coordinates, i.e., link length a_i, joint angle θ_i, link twist angle α_i, and link offset d_i. Fig. S.1 depicts a six-DoF HSR-JR680 industrial robot, whose initial values are recorded in Table S.II. According to [1], the relative pose of two adjacent links is determined through two rotation and translation motions, which obtains the transformation matrix

$$\mathbb{R}_i = \begin{bmatrix} \cos\theta_i & -\sin\theta_i\cos\alpha_i & \sin\theta_i\sin\alpha_i & a_i\cos\theta_i \\ \sin\theta_i & \cos\theta_i c\alpha_i & -\cos\theta_i s\alpha_i & a_i\sin\theta_i \\ 0 & \sin\alpha_i & \cos\alpha_i & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}. \tag{1}$$

Hence, a six-DoF robot kinematic model can be described as [19]

$$\mathbb{R} = \mathbb{R}_1\mathbb{R}_2\mathbb{R}_3\mathbb{R}_4\mathbb{R}_5\mathbb{R}_6 = \begin{bmatrix} h_x & h'_x & h''_x & S_x \\ h_y & h'_y & h''_y & S_y \\ h_z & h'_z & h''_z & S_z \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} H & S \\ 0 & 1 \end{bmatrix} \tag{2}$$

where H represents the nominal rotation matrix, and S is the nominal position vector [20]. Then, the pose error matrix is given as

$$\Delta\mathbb{R} = \mathbb{R}_r - \mathbb{R} \tag{3}$$

where $\Delta\mathbb{R}$, $\mathbb{R}_r$, and $\mathbb{R}$ are the pose deviation, actual pose matrix, and nominal pose matrix, respectively. Notably, the $\mathbb{R}_i$'s holomorphic differential form can be depicted as

$$d\mathbb{R}_i = \frac{\partial\mathbb{R}_i}{\partial\alpha_i}\Delta\alpha_i + \frac{\partial\mathbb{R}_i}{\partial a_i}\Delta a_i + \frac{\partial\mathbb{R}_i}{\partial d_i}\Delta d_i + \frac{\partial\mathbb{R}_i}{\partial\theta_i}\Delta\theta_i. \tag{4}$$

According to (2) and (4), the pose error model is given as

$$\Delta E = [A_1 \quad A_2 \quad A_3 \quad A_4]\begin{bmatrix}\Delta\alpha \\ \Delta a \\ \Delta d \\ \Delta\theta\end{bmatrix} = A\Delta\xi \tag{5}$$

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where A refers to the Jacobian matrix, $\Delta\xi$ denotes the kinematic parameter deviations including $\Delta\alpha$, Δa, Δd, and $\Delta\theta$.

B. Objective Function

According to the robot calibration principle [20] of a draw-string displacement sensor, the robot end-effector's position error is equivalent to the cable length error between the nominal and the actual, thus yielding the objective function

$$f = \min \|\delta E\| = \min \left[\frac{1}{2M} \sum_{i}^{M} \left(C_i - \hat{C}_i \right)^2 \right] \tag{6}$$

where C_i and $C\hat{}$ are the cable measured length and the cable nominal length, M is the sample count (initial measurement configuration count), and $C\hat{}$ is given as

$$\hat{C}_i = \sqrt{(S_i - S_0)^2} \tag{7}$$

where S_0 is a fixed point coordinate on the ground, and S_i refers to a nominal position coordinate of the robot end-effector. Note that according to the control principle of robots [20], operators can adjust the robot motion by inputting and adding the torsion angles $\{q_{i,1}, q_{i,2}, \ldots, q_{i,6}\}$ to the joint angles, thus yielding the S_i.

III. Kinematic Parameters Identification

A. MCS Method

To suppress the measurement noise and enhance the calibration performance, this article adopts an MCS method to optimize the robot calibration samples [16], [17]. For M different measuring configurations, we expand (5) as

$$\Delta E = \hat{A}\Delta\xi \tag{8}$$

where $A\hat{} = [A_1, A_2, \ldots, A_M]$ is an error extended Jacobian matrix for M initial configurations. According to the principle of SVD [16], [17], [18], (8) can be modified as

$$\Delta E = Z\Lambda\hat{Z}^T\Delta\xi \tag{9}$$

where Z and $Z\hat{}$ are the orthogonal matrices, Λ is a diagonal matrix including z singular values (σ_1, σ_2, $\cdots$, σ_z), given as

$$\Lambda = \begin{bmatrix} \sigma_1 & 0 & \cdots & 0 & 0 \\ 0 & \sigma_2 & \cdots & 0 & 0 \\ \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & \cdots & 0 & \sigma_z \\ 0 & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & \cdots & 0 & 0 \end{bmatrix} \tag{10}$$

where z is the dimension of the D-H parameter vector, σ_i ($i = 1$, 2, $\cdots$, z) refers to a singular value of the matrix $A\hat{}$. All singular values must satisfy the condition $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_z \geq 0$.

In the MCS, researchers have presented five observability indices for evaluating the sensitivity of the selected configuration on the identification results, i.e., $O_1 = M^{-0.5} \cdot (\sigma_1 \cdots \sigma_i \cdots \sigma_z)^{1/z}$, $O_2 = \sigma_z \cdot \sigma_{-11}$, $O_3 = \sigma_z$, $O_4 = \sigma 2\, z \cdot \sigma_{-11}$, $O_5 = \sigma - 11 + \cdots + \sigma - 1$ z. Based on these indices, another universal one (O_6) is developed for addressing measurement noise issue [16], which is given as

$$O_6 = \left[\left(\frac{1}{M} \sum_{i=1}^{M} \left(\frac{1}{\sigma_i^2} \right)^{\frac{u}{2}} \right) \right]^{-\frac{1}{u}} \tag{11}$$

where u is set to 2 in this work. Through the evaluation of observability indices, we adopt the DE algorithm to choose the optimized m measurement configurations.

B. AMPC Algorithm

Since the highly nonlinear nature of the D-H model, regularization terms are incorporated into (6) to prevent overfitting issues. Recently, immense research efforts have been made to develop diversified regularization terms, i.e., L_1 [21], L_2 [4], dropout [22], and log [23]. Based on the LM algorithm, this work adopts an L_2 regularization scheme to enhance the model stability, and the identification is given as

$$\begin{aligned} f = \frac{1}{2m} \sum_{i=1}^{m} \Bigg\| & C_i - \hat{C}_i - \frac{\partial \hat{C}_i}{\partial w} (w_{t+1} - w_t) \\ & - \frac{\partial \hat{C}_i}{\partial S_0} (S_{0,t+1} - S_{0,t}) \Bigg\|_2^2 \\ & + \frac{\lambda_0}{2} \left[(w_{t+1} - w_t)^2 + (S_{0,t+1} - S_{0,t})^2 \right] \end{aligned} \tag{12}$$

where w represents the D-H parameters including a, θ, α, d, w_t refers to the prior estimate vector, w_{t+1} represents the posted estimate. λ_0 is the regularization constant.

In order to build the planar constraint, the dial indicator is mounted vertically on the robot end-effector. The coordinates of a point on a plane that is solved by adding the dial indicator value to the nominal link length of the robot's sixth axis

$$\hat{a}_6 = a_6 + D - \hat{D} \tag{13}$$

where $a\hat{}_6$ and a_6 are the defined and the original link lengths of the robot's sixth axis, respectively, D refers to the length of the dial indicator, $D\hat{}$ denotes a dial indicator reading. Based on (1), (2), and (8), the measured points $C\hat{}_k$ on a plane are obtained. Then, the space plane constraint equation can be depicted as

$$\begin{aligned} & \Psi_k (w, G_k, \eta_k) = \left[\hat{C}_k - G_k \right]^T \eta_k = 0 \\ & \text{s.t.} \forall k \in \{1, 2, \ldots, n\} \end{aligned} \tag{14}$$

where ψ_k represents a constraint equation of the plane k, $\eta_k = (\eta_x, \eta_y, \eta_z)$ refers to a normal vector of the plane k, $G_k = (G_{k,x}, G_{k,y}, G_{k,z})$ denotes a point on the plane k, n total number of planes, which is set to 3. Aiming at building a model of space constraint minimization, we incorporate (14) into (12) to obtain

$$\begin{aligned} f_k = \frac{1}{2N} \sum_{k=1}^{n} \sum_{i=1}^{m} \Bigg\| & C_{i,k} - \hat{C}_{i,k} - \frac{\partial \hat{C}_{i,k}}{\partial w} (w_{t+1} - w_t) \\ & - \frac{\partial \hat{C}_{i,k}}{\partial S_0} (S_{0,t+1} - S_{0,t}) \Bigg\|_2^2 \\ & + \frac{\lambda_k}{2} \sum_{k=1}^{n} \left[(w_{t+1} - w_t)^2 + (S_{0,t+1} - S_{0,t})^2 \right] \\ \text{s.t.} & \left[\hat{\mathrm{C}}_\mathrm{k} - \mathrm{G}_\mathrm{k} \right]^\mathrm{T} \eta_\mathrm{k} = 0 \end{aligned} \tag{15}$$

where λ_k is the regularization constant for plane k.

ADMM algorithm can efficiently address constrained optimization issues [24]. Based on its principle, the Lagrange multiplier Υ_k is incorporated into (15) to build the augmented

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Lagrange function (16), shown at the bottom of this page, where ρ_k refers to a constraint factor for the plane k, $<\cdot>$ denotes the inner product between two matrices. Since the second derivative is omitted for simplicity, based on (15), the fixed point coordinate on the ground is fixed as a pretreatment

$$S_{0,t+1} = S_{0,t} + \left[\frac{1}{nm}\sum_{k=1}^{n}\sum_{i=1}^{m}\left(\frac{\partial \hat{C}_{i,k}}{\partial S_0}\left(\frac{\partial \hat{C}_{i,k}}{\partial S_0}\right)^T + \lambda_k I\right)\right]^{-1} \times \frac{1}{nm}\sum_{k=1}^{n}\sum_{i=1}^{m}\frac{\partial \hat{C}_{i,k}}{\partial S_0}\left(C_{i,k} - \hat{C}_{i,k}\right) \tag{17}$$

where I is a unit matrix. So $\eta_{k,t}$ is updated as

$$\eta_{k,t+1} = \eta_{k,t} - \left[\frac{1}{m}\sum_{i=1}^{m}\left(\rho_k \frac{\partial \Psi_{i,k}}{\partial \eta_k}\left(\frac{\partial \Psi_{i,k}}{\partial \eta_k}\right)^T\right) + \lambda_k I\right]^{-1} \times \frac{1}{m}\sum_{i=1}^{m}\left(\rho_k \Psi_{i,k}\frac{\partial \Psi_{i,k}}{\partial \eta_k} + \Upsilon_k \frac{\partial \Psi_{i,k}}{\partial \eta_k}\right). \tag{18}$$

Subsequently, $G_{k,t}$ is updated as

$$G_{k,t+1} = G_{k,t} - \left[\frac{1}{m}\sum_{i=1}^{m}\left(\rho_k \frac{\partial \Psi_{i,k}}{\partial G_k}\left(\frac{\partial \Psi_{i,k}}{\partial G_k}\right)^T\right) + \lambda_k I\right]^{-1} \times \frac{1}{m}\sum_{i=1}^{m}\left(\rho_j \Psi_{i,k}\frac{\partial \Psi_{i,k}}{\partial G_k} + \Gamma_j \frac{\partial \Psi_{i,k}}{\partial G_k}\right). \tag{19}$$

Notably, to address the issue of the comparability between various indicators, this work normalizes G_j and η_j to eliminate the dimensional impact

$$\eta_{k,t} = \frac{\eta_{k,t}}{\|\eta_{k,t}\|}, G_{k,t} = \frac{l_k G_{k,t}}{\|G_{k,t}\|} \tag{20}$$

where l_k represents the distance from the point on the plane k to the industrial robot coordinate origin. Afterwards, the iterative model of the kinematic parameter vector w_t is derived as

$$w_{t+1} = w_t + \frac{1}{nm}\sum_{k=1}^{n}\left[\sum_{i=1}^{m}\left(\frac{\partial \hat{C}_{i,k}}{\partial w}\left(\frac{\partial \hat{C}_{i,k}}{\partial w}\right)^T + \rho_k \frac{\partial \Psi_{i,k}}{\partial w}\left(\frac{\partial \Psi_{i,k}}{\partial w}\right)^T\right) + \lambda_k I\right]^{-1} \times \frac{1}{nm}\sum_{k=1}^{n}\sum_{i=1}^{m}\left[\frac{\partial \hat{L}_i}{\partial w}\left(C_{i,k} - \hat{C}_{i,k}\right) + \rho_k \Psi_{i,k}\frac{\partial \Psi_{i,k}}{\partial w} + \Upsilon_k \frac{\partial \Psi_{i,k}}{\partial w}\right]. \tag{21}$$

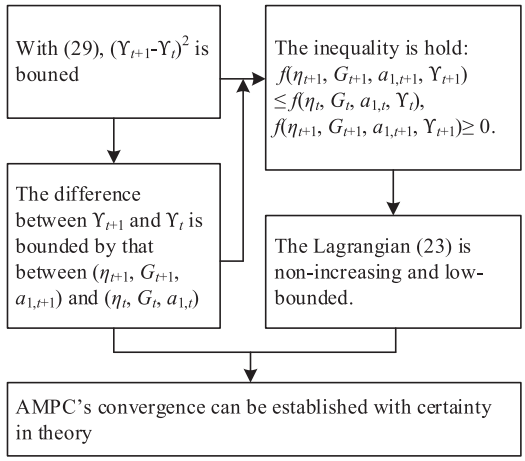


Fig. 1. Proof sketch of AMPC's convergence.

Finally, the rule of dual gradient ascent is adopted to update the Lagrange multiplier $\Gamma_{j,t+1}$

$$\Upsilon_{k,t+1} = \Upsilon_{k,t} + \frac{\tau_k \rho_k}{m}\sum_{i=1}^{m}\Psi_{i,k} \tag{22}$$

where τ_k is the step size of the Lagrangian multiplier at plane k.

C. Convergence Analysis of AMPC

This section provides a proof of convergence for the AMPC algorithm, with a schematic diagram depicted in Fig. 1. Due to the nonlinearity of the six-axis robot model, its computation process is immensely complex. To simplify the proof, we decouple the six-DoF D-H model into a two-DoF D-H model, taking a specific D-H parameter, such as the first link length a_1, as an example. Moreover, without affecting the convergence

$$\begin{aligned}
\arg\min_w f &= \arg\min_w\left(\frac{1}{n}\sum_{k=1}^{n} f_j\right) \\
&= \frac{1}{2nm}\sum_{k=1}^{n}\sum_{i=1}^{m}\left\|C_{i,k} - \hat{C}_{i,k} - \frac{\partial \hat{C}_{i,k}}{\partial w}(w_{t+1} - w_t) - \frac{\partial \hat{C}_{i,k}}{\partial S_0}(S_{0,t+1} - S_{0,t})\right\|_2^2 \\
&+ \frac{1}{nm}\sum_{k=1}^{n}\sum_{i=1}^{m}\left\langle\begin{pmatrix}\Psi_{i,k}(w, G_k, \gamma_k) + \frac{\partial \Psi_{i,k}}{\partial w}(w_{t+1} - w_t) \\ + \frac{\partial \Psi_{i,k}}{\partial G_k}(G_{k,t+1} - G_{k,t}) + \frac{\partial \Psi_{i,k}}{\partial \eta_k}(\eta_{k,t+1} - \eta_{k,t})\end{pmatrix}, \Upsilon_k\right\rangle \\
&+ \frac{1}{2nm}\sum_{k=1}^{n}\sum_{i=1}^{m}\rho_k\left\|\begin{matrix}\Psi_{i,k}(w, G_k, \gamma_k) + \frac{\partial \Psi_{i,k}}{\partial w}(w_{t+1} - w_t) \\ + \frac{\partial \Psi_{i,k}}{\partial G_k}(G_{k,t+1} - G_{k,t}) + \frac{\partial \Psi_{i,k}}{\partial \eta_k}(\eta_{k,t+1} - \eta_{k,t})\end{matrix}\right\|_2^2 \\
&+ \frac{1}{2n}\sum_{k=1}^{n}\lambda_k\left[(w_{t+1} - w_t)^2 + (S_{0,t+1} - S_{0,t})^2 + \sum_{k=1}^{n}\left((W_{k,t+1} - W_{j,t})^2 + (\gamma_{j,k+1} - \gamma_{k,t})^2\right)\right]
\end{aligned} \tag{16}$$

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of the AMPC algorithm, a single plane is considered for the convergence analysis of this algorithm.

Note that the end-effector position error and the cable length deviation between the nominal and the actual according to (6) and (7) have the same significance. To simplify the calculation, we consider utilizing the end-effector position error for convergence analysis. The regularization term is not an absolute factor affecting AMPC's convergence. So (16) is simplified as

$$f = \frac{1}{2m}\sum_{i=1}^{m}\left\|B_i - \hat{C}_i\right\|_2^2 + \frac{1}{m}\sum_{i=1}^{m}\left\langle \Psi_i\left(a_1, G, \eta\right), \Upsilon\right\rangle + \frac{1}{2m}\sum_{i=1}^{m}\rho\left\|\Psi_i\left(a_1, G, \eta\right)\right\|_2^2 \tag{23}$$

where a_1 is a elements of w, $B_i = (B_{i,x}, B_{i,y}, B_{i,z})$ is a measured position, $C_i = (C_{i,x}, C_{i,y}, C_{i,z})$ is a nominal position. Owing to the selected two-DOF robot, (2) can be described as $\mathbb{R} = \mathbb{R}_1\mathbb{R}_2$. Hence, based on (1), $C_{i,x}$, $C_{i,y}$, and $C_{i,z}$ are given as

$$\begin{cases} C_{i,x} = a_1 \times \cos\left(\theta_1 + q_{i,1}\right) + a_2 \times \cos\left(\theta_1 + q_{i,1}\right) \\ \quad \times \cos\left(\theta_2 + q_{i,2}\right) + d_2 \times \sin\left(\theta_1 + q_{i,1}\right) \times \sin\left(\alpha_1\right) \\ \quad - a_2 \times \sin\left(\theta_1 + q_{i,1}\right) \times \sin\left(\theta_2 + q_{i,2}\right) \times \cos\left(\alpha_1\right) \\ C_{i,y} = a_1 \times \sin\left(\theta_1 + q_{i,1}\right) + a_2 \times \cos\left(\theta_2 + q_{i,2}\right) \\ \quad \times \sin\left(\theta_1 + q_{i,1}\right) - d_2 \times \cos\left(\theta_1 + q_{i,1}\right) \times \sin\left(\alpha_1\right) \\ \quad + a_2 \times \cos\left(\theta_1 + q_{i,1}\right) \times \sin\left(\theta_2 + q_{i,2}\right) \times \cos\left(\alpha_1\right) \\ C_{i,z} = d_1 + d_2 \times \cos\left(\alpha_1\right) + a_2 \times \sin\left(\theta_2 + q_{i,2}\right) \times \sin\left(\alpha_1\right). \end{cases} \tag{24}$$

To be concise, three functions independent of a_1 are defined as

$$\begin{cases} \delta_1 = a_2 \times \cos\left(\theta_1 + q_{i,1}\right) \times \cos\left(\theta_2 + q_{i,2}\right) + d_2 \\ \quad \times \sin\left(\theta_1 + q_{i,1}\right) \times \sin\left(\alpha_1\right) \\ \quad - a_2 \times \sin\left(\theta_1 + q_{i,1}\right) \times sin\left(\theta_2 + q_{i,2}\right) \times \cos\left(\alpha_1\right) \\ \delta_2 = a_2 \times \cos\left(\theta_2 + q_{i,2}\right) \times \sin\left(\theta_1 + q_{i,1}\right) - d_2 \\ \quad \times \cos\left(\theta_1 + q_{i,1}\right) \times \sin\left(\alpha_1\right) \\ \quad + a_2 \times \cos\left(\theta_1 + q_{i,1}\right) \times \sin\left(\theta_2 + q_{i,2}\right) \times \cos\left(\alpha_1\right) \\ \delta_3 = d_1 + d_2 \times \cos\left(\alpha_1\right) + a_2 \times \sin\left(\theta_2 + q_{i,2}\right) \times \sin\left(\alpha_1\right). \end{cases} \tag{25}$$

Hence, $C_{i,x}$, $C_{i,y}$, and $C_{i,z}$ are redepicted as

$$\begin{aligned} C_{i,x} &= a_1 \times \cos\left(\theta_1\right) + \delta_1 \\ C_{i,y} &= a_1 \times \sin\left(\theta_1\right) + \delta_2 \\ C_{i,z} &= \delta_3. \end{aligned} \tag{26}$$

Following solution schemes of the AMPC algorithm, the learning rules of η, G, a_1, and Υ are sequentially updated as

$$\begin{cases} \eta_x = \frac{\sum_{i=1}^{m}\left(\Upsilon\kappa_{4,i} + \rho\kappa_{4,i}\left(\kappa_{5,i}\eta_y + \kappa_{6,i}\eta_z\right)\right)}{-\sum_{i=1}^{m}\left(\rho\kappa_{4,i}^2\right)} \\ \eta_y = \frac{\sum_{i=1}^{m}\left(\Upsilon\kappa_{5,i} + \rho\kappa_{5,i}\left(\kappa_{4,i}\eta_x + \kappa_{6,i}\eta_z\right)\right)}{-\sum_{i=1}^{m}\left(\rho\kappa_{5,i}^2\right)} \\ \eta_z = \frac{\sum_{i=1}^{m}\left(\Upsilon\kappa_{6,i} + \rho\kappa_{6,i}\left(\kappa_{4,i}\eta_x + \kappa_{5,i}\eta_z\right)\right)}{-\sum_{i=1}^{m}\left(\rho\kappa_{6,i}^2\right)} \end{cases} \tag{27a}$$

$$\begin{cases} G_x = \frac{\sum_{i=1}^{m}\left(\Upsilon + \rho\left(\eta_x a_1 \cos\theta_1 + \eta_x\delta_1 + \kappa_{5,i}\eta_y + \kappa_{6,i}\eta_z\right)\right)}{\sum_{i=1}^{m}\left(\rho\eta_x\right)} \\ G_y = \frac{\sum_{i=1}^{m}\left(\Upsilon + \rho\left(\eta_y a_1 \sin\theta_1 + \eta_y\delta_2 + \kappa_{4,i}\eta_x + \kappa_{6,i}\eta_z\right)\right)}{\sum_{i=1}^{m}\left(\rho\eta_y\right)} \\ G_z = \frac{\sum_{i=1}^{m}\left(\Upsilon + \rho\left(\eta_z\delta_3 + \kappa_{4,i}\eta_x + \kappa_{5,i}\eta_y\right)\right)}{\sum_{i=1}^{m}\left(\rho\eta_z\right)} \end{cases} \tag{27b}$$

$$a_1 = \frac{\sum_{i=1}^{m}\left(\kappa_{1,i} + \Upsilon\kappa_{2,i} - \rho\kappa_{2,i}\kappa_{3,i}\right)}{-\sum_{i=1}^{m}\left(1 + \rho\kappa_{2,i}^2\right)} \tag{27c}$$

$$\Upsilon = \Upsilon + \frac{\tau\rho}{m}\sum_{i=1}^{m}\left(\kappa_{4,i}\eta_x + \kappa_{5,i}\eta_y + \kappa_{6,i}\eta_z\right) \tag{27d}$$

where $\kappa_{1,i}$–$\kappa_{6,i}$ are defined as

$$\begin{cases} \kappa_{1,i} = \delta_1\cos\left(\theta_1 + q_{i,1}\right) + \delta_2\sin\left(\theta_1 + q_{i,1}\right) \\ \quad - B_{i,x}\cos\left(\theta_1 + q_{i,1}\right) - B_{i,y}\sin\left(\theta_1 + q_{i,1}\right) \\ \kappa_{2,i} = \eta_x\cos\left(\theta_1 + q_{i,1}\right) + \eta_y\sin\left(\theta_1 + q_{i,1}\right) \\ \kappa_{3,i} = \eta_x\left(\delta_1 - G_x\right) + \eta_y\left(\delta_2 - G_y\right) + \eta_z\left(\delta_3 - G_z\right) \\ \kappa_{4,i} = a_1\cos\left(\theta_1 + q_{i,1}\right) + \delta_1 + G_x \\ \kappa_{5,i} = a_1\sin\left(\theta_1 + q_{i,1}\right) + \delta_2 + G_y \\ \kappa_{6,i} = \delta_3 + G_z. \end{cases} \tag{28}$$

Based on the ADMM principle [24], we design a learning sequence comprising four subtasks, each resolved by incorporating the latest solutions to previously addressed tasks and considering all relevant entries. Note that η, G and a_1 are updated by (27a), (27b), and (27c), i.e., the formula serial numbers in (29a), (29b), and (29c); Υ is updated via dual gradient ascent (27b) in Job Four, respectively. Hence, in $(t+1)$th iteration, these subtasks can be depicted as follows:

Subtask One:

$$\eta_{t+1} \overset{(27a)}{\leftarrow} \arg\min_{\eta} f\left(\eta, G_t, a_{1,t}, \Upsilon_t\right) \tag{29a}$$

Subtask Two:

$$G_{t+1} \overset{(27b)}{\leftarrow} \arg\min_{G} f\left(\eta_{t+1}, G, a_{1,t}, \Upsilon_t\right) \tag{29b}$$

Subtask Three:

$$a_{1,t+1} \overset{(27c)}{\leftarrow} \arg\min_{a_1} f\left(G_{t+1}, \eta_{t+1}, a_1, \Upsilon_t\right) \tag{29c}$$

Subtask Four:

$$\Upsilon_{t+1} \overset{(27d)}{\leftarrow} \Upsilon_t + \frac{\tau\rho}{m}\frac{\partial}{\partial\Upsilon} f\left(G_{t+1}, \eta_{t+1}, a_{1,t+1}, \Upsilon_t\right). \tag{29d}$$

Note that the formula serial number on the arrow represents the updated formula. For instance, the (29a) for updating W follows the computation rules in (27a). With (29), they contain the following principles.

1) To update the variables of η, G, a_1, and Υ, each subtask is solved by focusing on a variable and alternately keeping the others fixed, which makes the related subtasks solvable analytically.
2) Each subtask fully utilizes all the entries.

This design endows the AMPC algorithm with the potential for rapid convergence, and the proof of its convergence involves two steps.

Step 1: The difference between Υ_{t+1} and Υ_t is bounded by that between $(\eta_{t+1}, G_{t+1}, a_{1,t+1})$ and $(\eta_t, G_t, a_{1,t})$.
Step 2: (23) is nonincreasing and lower bounded.

The detailed proofs are provided in the SF[2].

D. Algorithm Design and Analysis

The MCS-AMPC calibrator for the robot calibration is designed in Algorithm I, where Algorithm II is a submodule of Algorithm I for calculating O_6. Note that we provide the detailed Algorithm I and Algorithm II in the SF[2]. Algorithm I consists of two main steps. The first step is the MCS for the best group of samples, and the second step is the AMPC for the kinematic parameter identification. The cost has three factors as follows.

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1) Initialization:

$$T_1 = \Theta(r_0) \tag{30}$$

where r_0 is the dimension of w_0.

2) Time cost of the MCS step

$$T_2 = \Theta\left[\left(\sum_{t=z}^{M} t\right) + K_1 \times m\right]. \tag{31}$$

3) Time cost of the AMPC step

$$T_3 = \Theta\left[(K_2 + K_3) \times m\right]. \tag{32}$$

According to the abovementioned discussion, it can extrapolate the condition $r_0 \ll (K_2{+}K_3) \times m$, and $T_1 \ll \Theta[(K_2{+}L_3) \times m]$. Hence, the time cost of the MCS-AMPC calibrator can be expressed as

$$T_K = T_1 + T_2 + T_3 \approx \Theta\left(\left(\sum_{t=z}^{M} t\right) + (K_1 + K_2 + K_3) \times m\right) \tag{33}$$

where w_0 refers to the initial D-H parameters, K_1 is the iteration count for the DE algorithm, K_2 is the iteration count for computing S_0, and K_3 is the iteration count for calculating G_k, η_k, w_k, and Υ_k, $G_{0,k}$, $\eta_{0,k}$, and $\Upsilon_{0,k}$ are the initial values.

IV. Experiments and Results

Part of the figures regarding experiments are also shared in the SF available online[2].

A. General Settings

Evaluation Metrics: this study adopts three evaluation metrics, i.e., root-mean-square error (RMSE), mean error (MEAN), and maximum error (MAX) [1], [2], [3], [4], [5]

$$\text{RMSE} = \sqrt{\frac{1}{m}\sum_{i=1}^{m}\left(C_i - \hat{C}_i\right)^2}, i = 1, 2, \ldots, m$$

$$\text{MEAN} = \frac{1}{m}\sum_{i=1}^{m}\left|C_i - \hat{C}_i\right|$$

$$\text{MAX} = \max\left|C_i - \hat{C}_i\right|.$$

Datasets: To verify the performance of the MCS-AMPC calibrator, this work establishes three datasets based on three different planes of an HSR-JR680 industrial robot.

1) D1: HSR-P1. It is collected on a plane that is parallel to the ground, containing 800 samples shown in Fig. S.2(a).
2) D2: HSR-P2. As depicted in Fig. S.2(b), 800 samples have been collected based on a plane that forms a 45-degree angle with the ground.
3) D3: HRS-P3. This dataset is built upon a plane perpendicular to the ground comprising 800 samples, as in Fig. S.2(c).

Moreover, to verify the effectiveness of the proposed calibration method in actual scenarios, this study inputs the compensated D-H parameter values into the HSR-JR680 industrial robot and remeasures 200 samples in its workspace, forming the subdataset HRS-AC. The mentioned datasets have been combined into HRS-P and are available on the website[1]. As listed in Table I, each sample on HRS-P consists of an actual cable length measured by a drawstring displacement sensor, six joint angles, and a dial indicator reading.

TABLE I
Detailed Five Samples

No.	$q1$/°	$q2$/°	$q3$/°	$q4$/°	$q5$/°	$q6$/°	L/mm	d/mm
1	2.79	−46.56	171.15	−358.27	54.17	21.80	315.83	6.75
2	2.81	−47.72	173.05	−358.25	53.43	21.80	305.64	5.83
3	2.85	−48.87	174.92	−358.24	52.71	21.80	296.25	5.02
4	2.85	−47.27	172.32	−358.26	53.72	21.80	309.70	6.15
5	2.77	−45.88	170.03	−358.28	54.61	21.80	322.24	7.45

This study adopts a data-driven approach for conducting experiments. Considering measurement noise and computational efficiency, a systematic random sampling method is first adopted to randomly select varying amounts of data in D1-3 for robot calibration. Then, based on the above results, we ascertain the data quantity needed for each dataset, thereby forming three smaller calibration datasets ($\hat{\text{D}}$1-3). In order to investigate the effect of multiplanar constraints, this article establishes P1-3 through $\hat{\text{D}}$1-3, specifically, P1: $\hat{\text{D}}$1, P2: $\hat{\text{D}}$1-2, and P3: $\hat{\text{D}}$1-3. Note that we employ the MCS method to further exclude data points that are either outliers or lack overall representativeness. Moreover, each experimental dataset is split into an 80% training dataset and a 20% test dataset. To mitigate the bias due to data splitting, each experiment is conducted ten times and the standard deviation is reported.

Experimental Setup. Fig. S.3(a) shows the experimental system of this work, i.e., a drawstring displacement indicator, an HSR-JR680 industrial robot, a computer, a drawstring displacement sensor, a high-precision marble gauge block, and a dial indicator. Fig. S.3(b)–(c) depicts the working interface of the data acquisition software (LabVIEW), on the two different samples. Moreover, Tables S.II-V provide detailed information about the aforementioned devices, which satisfy the hardware requirements for robot calibration, featuring low theoretical measurement noise.

Experimental Process: From Fig. S.3(a), the experiments are conducted in an actual scenario with a stable operating environment, including controlled temperature and humidity, and a reliable power supply, which helps reduce theoretical measurement noise. The experimental process can be divided into the following steps.

Step 1: Experimental preparation. First, firmly attach the dial indicator to the robot's end flange and place the drawstring displacement sensor at a fixed point on the desk. Connect the other end of the drawstring to the robot's end-effector.

Step 2: Data collection. Subsequently, precisely control the robot to move along the predetermined trajectory to each preset point, recording the necessary data at each point (including a micrometer reading, a wire length, and six robot torsion angles). Note that the end of the dial indicator must touch the marble plane when the robot reaches the preset point. Fig. S.2(a)–(c) depicts the sample collection process on three different planes corresponding to D1-3. For the sample collection process at various measurement positions, this work takes D1 as an example, as shown in Fig. S.4(a)–(c). The data acquisition software records the cable lengths and the dial indicator readings in real-time.

Step 3: Calculation process. Fig. S.5 shows the calculation process of robot calibration. First, ascertain the quantity of data

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needed for D1-3 based on the systematic random sampling method, forming three calibration datasets. Afterwards, adopt the MCS method to process these samples and identify the kinematic parameters with the proposed AMPC algorithm. Note that the detailed computational steps for MCS-AMPC can also be found in its algorithm pseudocode. Then, an ensemble is established based on different solutions to obtain a stable solution. Finally, the kinematic errors are compensated to improve the industrial robot's accuracy.

Note that the training process of all the calibrators has a common termination condition, either the validation error is less than 10^{-3}mm for five consecutive iterations, or the training iteration counts surpass the preset limit of 200 iterations.

B. Sample Sizes

For verifying the impact of the measurement noise, we explore the calibration performance of the AMPC algorithm on D1-3 with different sample sizes (100, 200, 300, 500, and 700), labeled as S100, S200, S300, S500, and S700, respectively. Table S.VI and Table S.VII, respectively, list the calibration accuracy and total time costs of the proposed AMPC algorithm for different sample sizes on D1-3. From the above results, we find that as follows.

1) *Increasing the data volume is beneficial for reducing measurement noise:* For instance, from Table S.VI, on D1, it is seen that S200 achieves RMSE, MEAN, and MAX at 0.594 mm, 0.492 mm, and 0.965 mm, which are 4.65%, 5.92%, and 3.98% lower than those of S100. However, S200's RMSE is only 0.34%, 0.51%, and 0.67% higher than those achieved by S300, S500, and S700, respectively. For the MEAN, the outputs by S300, S500, and S700 are 0.490 mm, 0.489 mm, and 0.488 mm, respectively. Hence, S200 is also slightly higher than its peers. Meanwhile, the MAX for S200 exceeds S300's 0.963 mm by just 0.21%, S500's 0.961 mm by 0.42%, and S700's 0.960 mm by 0.52%. On D2-3, similar findings are noted, as depicted in Table S.VI. Therefore, increasing the data volume can improve calibration accuracy and reduce measurement noise to some extent.
2) *Higher data volume causes a drop in computational efficiency:* For instance, as shown in Table S.VII, S200 consumes 22.7 s to achieve RMSE convergence on D1, which is 22.53%, 44.77%, and 57.49% faster than S300's 29.3 s, S500's 41.1 s, and S700's 53.4 s, respectively. In contrast, S100 takes 21.3 s to converge in RMSE, which is only 1.4 s less than S200.

On D2-3, similar conclusions are observed, as listed in Table S.VII. Thus, high data volumes can adversely affect computational efficiency.

Based on the abovementioned analysis, increasing data volume is beneficial for noise reduction. Nevertheless, when the data volume surpasses a certain limit, its effect on measurement noise reduction lessens and computational efficiency decreases. Hence, considering measurement noise and computational efficiency, S200 achieves satisfactory calibration performance on D1-3. Therefore, this study randomly selects 200 samples from D1-3 as the calibration datasets ($\hat{\text{D}}$1-3).

C. Comparisons of All Calibrators

Compared Calibrators: To evaluate the calibration performance of the MCS-AMPC calibrator, we perform extensive comparison experiments with eight state-of-the-art algorithms on P1-3. All adopted algorithms are listed as follows.

1) M1: A PSO method with MPC algorithm, which enables fast convergence [8], [20].
2) M2: An extended Kalman filter with MPC algorithm, which can effectively address non-Gauss noise [2], [3].
3) M3: A LM with MPC algorithm [1].
4) M4: A radial basis function neural network with MPC algorithm [25].
5) M5: An unscented Kalman filter with MPC algorithm, which can effectively nonlinear issues [26].
6) M6: A LS with MPC algorithm, which is the most prevalent in robot calibration studies [9].
7) M7: A particle filter with MPC algorithm, which can reduce noises in the process of the measurement [2].
8) M8: A single AMPC algorithm presented in Section III-B.
9) M9: An MCS-AMPC calibrator proposed in Section III.

Compared results: On $\hat{\text{D}}$1-3, Fig. S.6 shows how the number of measurement configurations affects O_6. Fig. S.7 depicts the optimal selection process of measurement configurations. Fig. S.8 illustrates the distributions of the observability indices, respectively. Fig. S.9 depicts calibration accuracy and total time cost of all calibrators on P1-3. Fig. S.10 illustrates the convergence curves of all calibrators on P1-3. Fig. S.11 shows the position errors of the sample points (before calibration (BC), M2-4, M7, and M9) on P1-3. Note that due to the distinct differences in positioning error distribution between M2-4, M7, and M9, this study selects these representative calibrators to compare the positioning errors, aiming to improve the readability of Fig. S.11(d)–(f). Table S.VIII records the calibration accuracy of M1-9 on P1-3. Table S.IX shows the time costs of M1-9 on P1-3. On the basis of them, we have the following findings.

1) *AMPC achieves higher calibration accuracy of industrial robots than its peers:* From Fig. S.9(a)–(c), it is evident that an AMPC algorithm, i.e., M8 consistently outperforms the M1-7 in terms of calibration accuracy. For instance, as listed in Table S.VIII, on P1, M8's RMSE is 0.594 mm, which is about 39.33% lower than M1's 0.979 mm, 52.56% lower than M2's 1.252 mm, 9.03% lower than M3's 0.653 mm, 23.06% lower than M4's 0.772 mm, 44.38% lower than M5's 1.068 mm, 14.90% lower than M6's 0.698 mm, and 46.53% lower than M7's 1.111 mm. For the MAEN, M8 achieves 0.493 mm, which is about 43.96%, 57.66%, 14.14%, 26.89%, 44.53%, 17.58%, and 51.38% lower than M1-7's. Considering the MAX, they are 1.755 mm, 2.425 mm, 1.125 mm, 1.522 mm, 1.858 mm, 1.321 mm, and 2.101 mm in M1-7, respectively, and M8 also has a lower Max than its peers. Similar results are encountered in P2-3, as summarized in Table S.VIII and Fig. S.9(a)–(c). Hence, AMPC outperforms the comparison calibrators in robot calibration accuracy.
2) *MCS is instrumental in addressing the measurement noise issue:* For instance, As shown in Fig. S.6, the observability

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indices increase with the number of measurement configurations on $\hat{\mathrm{D}}$1-3 and then level off at about 100 samples. According to [16], the larger observability index indicates a more stable kinematic parameters identification. Thus, appropriate measurement configuration numbers are all selected as $m = 100$ on $\hat{\mathrm{D}}$1-3. Moreover, from Fig. S.7, the $-O_6$ of $\hat{\mathrm{D}}$1-3 converge to -3.73×10^{-15}, -3.16×10^{-15}, and -2.89×10^{-15}, respectively. Then, the 100 samples at the $-O_6$ convergence point (i.e., the observability index is the maximum value) are obtained as the optimal measurement configurations for the following kinematic parameter identification process. Afterwards, to verify this result, this work randomly selects 100 measurement configurations from the samples of $\hat{\mathrm{D}}$1-3 and calculates the corresponding observability indices, respectively. By repeating the above process 1000 times, the distribution of the O_6 values is obtained, as depicted in Fig. S.8. For instance, in Fig. S.8(a), on $\hat{\mathrm{D}}$1, the values of O_6 in these 1000 repetitions are not larger than 3.73×10^{-15}, which demonstrates that the selection of the optimal measurement configurations is effective. As depicted in Fig. S.8(b)–(c), similar phenomena are obtained on $\hat{\mathrm{D}}$2-3.

After conducting the MCS process, the optimal measurement configurations are adopted to calibrate the kinematic parameters. From Fig. S.9(a)–(c) and Table S.VIII, it is evident that the MCS-AMPC calibrator, i.e., M9 outperforms the M1-8. As listed in Table S.VIII, M9's RMSE, Mean, and MAX on P1 are 0.594 mm, 0.449 mm, and 0.885 mm, which are about 7.58%, 8.73%, and 8.29% lower than that of M8. On P2-3, similar results can be obtained, as illustrated in Table S.VIII. Hence, MCS can effectively reduce the measurement noise.

1) *AMPC has a rapid convergence rate:* For instance, as shown in Fig. S.10 and Table S.IX, on P1, M1-7, respectively, take 26, 12, 50, 85, 16, 65, and 13 iterations to converge in RMSE. In contrast, M8 only consumes 11 iterations in RMSE, thus obtaining the lowest iteration count among its peers, which is attributed to the fact that the Lagrange multiplier term in the AMPC addresses long tail convergence issues effectively. From Table S.IX, similar results are also obtained on P2-3.
2) *Robot calibration accuracy benefits from the MPC strategy:* For instance, from Fig. S.9(a)–(c) and Table S.VIII, M9's RMSE, MEAN, and MAX are 0.502 mm, 0.411 mm, and 0.815 mm on P3, which are about 8.56%, 8.46%, and 7.91% lower than M9's 0.549 mm, 0.449 mm, and 0.885 mm on P1, and 4.92%, 2.84%, and 1.69% lower than M9's 0.528 mm, 0.423 mm, and 0.829 mm on P2. For the M1-8, similar outcomes are also obtained, as illustrated in Fig. S.9(a)–(c) and Table S.VIII. Thus, the above results demonstrate the contribution of the MPC strategy to calibration accuracy.
3) *MCS-AMPC achieves a competitive computational efficiency:* For instance, from Table S.IX and Fig. S.10(c), M9 takes 62.8 s to converge in RMSE on P3, which is 31.07% of M1's 202.1 s, 96.32% of M2's 65.2 s, 21.71% of M3's 289.3 s, 52.25% of M4's 120.2 s, 76.43% of M5's 82.2 s, 19.11% of M6's 328.6 s, 98.74% of M7's 63.6 s, and 96.17% of M8's 65.3 s. The reason for this phenomenon is that M9 enables parallel computation for the MCS process of multiple planar data, as well as the reduced amount of data trained by AMPC after MCS. Therefore, AMPC exhibits a fast convergence rate, which contributes to a reduction in the time cost.
4) *The Wilcoxon signed-rank test validates the performance benefits of MCS-AMPC in robot calibration:* To check whether the MCS-AMPC outperforms several state-of-the-art algorithms in accuracy, and analyze the effectiveness of the proposed MCS-AMPC in calibration accuracy, Wilcoxon signed ranks are adopted to test Table S.VIII. Note that it contains three factors (R+, R-, and p-value). The higher R+ and lower R- indicate more reliable accuracy, and the p-value represents the significance level with an accepted level of 0.05. From Table S.X, comparing M9 with M1-8, the resultant p-values are all 0.002, which is far smaller than 0.05. Moreover, $\mathrm{R}+ = 45$ and $\mathrm{R}- = 0$ demonstrate high significance. Consequently, the accuracy gain in robot calibration achieved by the MCS-AMPC calibrator surpasses that of its peers.
5) *MCS-AMPC lowers the position error of the industrial robot:* According to the abovementioned analyses, this article adopts M9 to calibrate the kinematic parameters of an industrial robot. In total, 30 points were selected from the training samples to compare the robot's positioning error after calibration using various algorithms on P1-3. As shown in Fig. S.11(a)–(c), the position errors are decreased after calibration on P1-3. In addition, M9 has the smallest error among all algorithms, as depicted in Fig. S.11(d)–(f) on P1-3. Accordingly, the MCS-AMPC calibrator, i.e., M9, is practically feasible for addressing the robot position error issues.

D. Ensemble of Kinematic Parameter Vectors

Due to the linear correlation between D-H parameters and the involvement of nonlinear equations and multivariable complex relationships during optimization, robot calibration can be characterized as a nonconvex optimization problem. This issue frequently has multiple local optimal solutions, with different solutions potentially corresponding to different robot configurations, thereby impacting accuracy and performance. To obtain an efficient and stable solution, this article designs an ensemble of different solutions. Specifically, on P3, M9 is adopted to calibrate the robot's kinematic parameters and collect several different solutions. Then, these solutions are integrated through the typical AdaBoost algorithm. Table S.XI records seven D-H parameter vectors (DH1-7) on P3 after calibration by M9. Table S.XII shows the calibration accuracy of DH1-7 on P3. Based on DH1-7, we develop seven ensembles, detailed as follows.

1) E1: It integrates DH1 and DH2 vectors of the D-H parameters.
2) E2: It consists of DH1, DH2, and DH3 vectors of the D-H parameters.
3) E3: It encompasses four vectors of the D-H parameters, from DH1 to DH4.
4) E4: It includes five vectors of the D-H parameters, from DH1 to DH5.

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5) E5: It consists of six vectors of the D-H parameters, ranging from DH1 to DH6.
6) E6: It contains seven vectors of the D-H parameters, from DH1 to DH7.

Table S.XIII illustrates the calibration accuracy of E1-7 on P3. Table S.XIV lists the D-H parameter deviations of the E7 ensemble on P3. From the aforementioned results, we summarize that as follows.

1) *The calibration accuracy of different solutions varies slightly:* For instance, as shown in Table S.XII, DH4 achieves the RMSE at 0.499 mm, which is 0.60% lower than DH1's 0.502 mm, 0.40% lower than DH2's 0.501 mm, 0.80% lower than DH3's 0.503 mm, 0.20% lower than DH5's 0.5 mm, 0.99% lower than DH6's 0.504 mm. Meanwhile, DH4's MEAN is 0.408 mm, which is 0.73%, 0.24%, 0.97%, 0.49%, 1.21%, and 1.45% lower than DH1's 0.411 mm, DH2's 0.409 mm, DH3's 0.412 mm, DH5's 0.41 mm, DH6's 0.413 mm, and DH7's 0.414 mm, respectively. Moreover, DH4's MAX is 0.408 mm, which is 0.37%, 0.12%, 0.49%, 0.25%, 0.61%, and 0.73% lower than that of DH1-3 and DH5-7, respectively. Hence, different solutions affect the robot's calibration accuracy. This phenomenon can be explained by the fact that solutions derived from different convergence states correspond to different robot configurations, leading to variations in accuracy. However, the accuracy variation among different solutions is minimal, less than 1.2%, indicating that the proposed MCS-AMPC algorithm is stable in addressing the issue of robot calibration.
2) *The ensemble facilitates the determination of the final solution:* As illustrated in Table S.XIII, on P3, it is observed that the calibration accuracy initially increases and then stabilizes as the number of solutions increases. Specifically, E5's RMSE is 0.488 mm, which is 1.81% lower than E1's 0.497 mm, 1.21% lower than E2's 0.494 mm, 0.61% lower than E3's 0.491 mm, and 0.20% lower than E4's 0.489 mm. Considering MEAE, the outputs of E1-5 are 0.407 mm, 0.404 mm, 0.402 mm, 0.401 mm, and 0.4 mm. E5's MAX is 0.8 mm, which is 1.23%, 0.74%, 0.38%, and 0.12% lower than that of E1-4, respectively. Moreover, E6 exhibits the same calibration accuracy as E5. Therefore, the ensemble of various solutions can further enhance the robot's calibration accuracy. Based on the analysis above, this article identifies E5 as the final solution for compensating the kinematic parameters of the HRS-JR680 industrial robot, as shown in Table S.XIV.
3) *After D-H parameter compensation, the robot's position accuracy is effectively improved:* Based on the above analyses, this study inputs E5 into the HRS-JR680 industrial robot for D-H parameter compensation and then obtains the subdataset HRS-AC after recollecting samples. We choose 30 samples from HRS-AC to compare the robot's positioning accuracy before and after D-H parameter compensation. From Fig. S.12, the robot's position accuracy is effectively enhanced after compensating the D-H parameters. Thus, E5, i.e., the selected D-H parameter vector, proves to be a practical solution in actual scenarios.

V. Conclusion

To establish a more precise calibration of industrial robots within a confined workspace, this work presented a new MCS-AMPC calibrator, which employs an MCS method for eliminating measurement noise and identifies the kinematic parameter errors through an AMPC algorithm. Note that the theoretical convergence of the AMPC algorithm is conclusively proved. Finally, an ensemble was constructed from several local optimal solutions, resulting in the final solution. Extensive empirical studies demonstrated that the proposed MCS-AMPC calibrator can reduce the mean error from 2.45 mm to 0.411 mm, and the ensemble can further enhance robot calibration accuracy by 2.68%. Consequently, the MCS-AMPC calibrator provides a practical solution for researchers and engineers involved in robot calibration. In the future, we plan to address the following issues.

1) To further improve the positioning accuracy of the robot, nongeometric errors should not be ignored, even though they only account for less than 10% of the positioning error.
2) An online robot calibration scheme is intriguing, as it can mitigate the impact of offline calibration interruption on industrial production progress.
3) On account of the same calibration process, the MCS-AMPC calibrator can also achieve calibration for different industrial robots by simply replacing the initial kinematic parameters and measurement data. Moreover, in actual factory conditions, the complex working environment and long-term wear and tear of the robot can reduce its accuracy. Consequently, to verify the generalization and industrial applicability of the MCS-AMPC calibrator, it will be interesting to implement it on various industrial robots in different production environments.
4) Several application-oriented verification methods, e.g., multiview measurement accuracy comparison of point cloud reconstruction and robot motion accuracy comparison, can be utilized to evaluate the high accuracy and reliability levels that robots achieve in actual applications.
5) Efficient adaptive hyperparameter tuning methods can improve algorithm performance, accuracy, and generalization. Hence, it is highly interesting to develop applicable adaptive hyperparameter tuning methods to further enhance the calibration performance of both the compared algorithms and the proposed AMPC algorithm.
6) The ADMM algorithm can address various types of constraints. Considering the different constraint spaces for different robots, the proposed method can employ various high-precision gauge blocks for data collection and replace the corresponding constraints, e.g., spherical constraint, linear constraint, and distance constraint in (16) to achieve robot calibration.

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